

DIYA NILESH DESAI

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PROFESSIONAL SUMMARY

Recent B.Sc. Data Science graduate (CGPA: 8.16/10) with hands-on experience in machine learning, deep learning, computer vision, Generative AI, time-series forecasting, and business analytics. Built end-to-end AI applications including a RAG-based document Q&A system using LangChain, HuggingFace, and ChromaDB. Proficient in Python, Scikit-learn, NumPy, Pandas, SQL, Power BI, and Tableau. Backed by professional Python internship experience, multiple real-world projects, and certifications across AI, ML, and cloud platforms. A fast learner with strong analytical thinking and a passion for building intelligent, data-driven solutions.

EDUCATION

B.Sc. – Data Science

Graduate – May 2026

Navrachana University, Vadodara | CGPA: 8.16 / 10

INTERNSHIP EXPERIENCE

Python Developer Intern | BrainlyBeam InfoTech Pvt. Ltd.

January 2026 – May 2026

- Developed and maintained Python scripts supporting data collection, processing, and transformation workflows in a professional team environment.
- Applied data cleaning, validation, and preprocessing techniques on structured datasets to ensure quality and consistency for downstream use.
- Collaborated with senior developers on project workflows; managed version control using Git/GitHub and documented processes and methodologies clearly.
- Gained practical exposure to agile team collaboration, real-world development cycles, and cross-functional problem-solving in a fast-paced environment.

TECHNICAL SKILLS

Programming	Python, SQL, R
Generative AI	RAG (Retrieval Augmented Generation), LangChain, HuggingFace, LLM APIs (Groq/Llama), ChromaDB, Vector Databases, Prompt Engineering, FastAPI
ML & AI	Supervised & Unsupervised Learning, Regression, Classification, Clustering, Model Evaluation, Feature Engineering, Predictive Modelling
Deep Learning	CNN, YOLOv8, OpenCV, Image Annotation, Data Augmentation, Real-time Inference
Libraries	NumPy, Pandas, Scikit-learn, Matplotlib
Forecasting	Time Series Analysis, ARIMA, Holt-Winters, Demand Forecasting
Data Analysis	Statistical Analysis, EDA, Data Cleaning, Data Validation, Data Preprocessing, Excel (Pivot Tables, VLOOKUP, INDEX/MATCH)
Visualization	Power BI (DAX, KPI Dashboards), Tableau, Matplotlib, Streamlit
Cloud & Tools	AWS (Cloud Foundations, Data Engineering, ML Foundations), Git, GitHub, Roboflow

PROJECTS

Petal AI — Document Q&A Study Assistant | Python, FastAPI, LangChain, HuggingFace, ChromaDB, Groq LLM

- Built a full-stack RAG application enabling conversational Q&A over any uploaded PDF document with accurate, hallucination-free answers grounded in document context.
- Implemented end-to-end AI pipeline covering PDF parsing, text chunking, HuggingFace embedding generation, ChromaDB vector storage, and semantic similarity search.
- Integrated LangChain conversational memory to resolve follow-up questions using chat history, and Groq Llama 3.3 70B for context-grounded answer generation.
- Built a FastAPI REST backend with upload and ask endpoints alongside a custom responsive front-end chat UI.

Driver Drowsiness Detection | Python, CNN, OpenCV, Machine Learning

- Designed and trained a CNN model on facial and eye-region datasets to classify driver states in real time from live video streams.
- Integrated Eye Aspect Ratio (EAR) and facial landmark detection to improve model robustness and prediction accuracy.

Smart Parking System – Computer Vision | *Python, YOLOv8, OpenCV* | **Hackathon 2nd Runner-Up**

- Trained a YOLOv8 object detection model on custom-annotated datasets to automate parking occupancy classification from image and video feeds.
- Built a complete end-to-end ML pipeline from data annotation and augmentation through model training, evaluation, and inference; awarded 2nd Runner-Up at Indigenous Hackathon, March 2025.

Forecasting Renewable Energy Capacity | *Python, ARIMA, Holt-Winters, Streamlit*

- Analysed multi-year historical datasets to identify growth trends and seasonal patterns; built ARIMA and Holt-Winters models for state-wise solar and wind capacity prediction over 2 to 3 years.
- Developed an interactive Streamlit application for scenario-based forecast comparison and stakeholder reporting.

Nutritional Health Analysis Dashboard | *Power BI, DAX, Excel*

- Collected and structured multi-source retail datasets; built interactive Power BI dashboards with custom DAX measures to track category-wise and brand-wise KPIs for business decision-making.

AWARDS

Indigenous Hackathon – 2nd Runner-Up | March 2025 | Smart Parking System using Computer Vision

CERTIFICATIONS

- AWS Academy – Cloud Foundations | Data Engineering | Machine Learning Foundations
- AWS – Machine Learning for Natural Language Processing
- Deloitte Australia – Data Analytics Job Simulation
- Tata – Data Visualization: Empowering Business with Effective Insights
- IBM – Getting Started with Data